

第 16360-C 章

單元變電站

第一部份 通則

1.1 說明

- A. 本部份涵蓋單元變電站及零配件之設計、供料、安裝及測試。
- B. 單元變電站主要包含LV Voltage Switchboard及Cast Resin Encapsulated Dry Transformer (480V)。
- C. 主要的工程範圍如下述，以下列出主要項目工程(但為完成整個工程並未限定僅列出之項目)。
 - 1. 樹脂模鑄型乾式變壓器外箱及低壓電盤等設備提供及安裝，溫度電驛電源須使用DC電源，且含DC Charger及Batory的設備提供及安裝，電池容量承商須依實際之需求，並按照IEEE規範提出計算書送審。
 - 2. 本案低壓盤採用標準型試驗低壓開關盤，盤內隔離方式須符合IEC 60439-1 FORM 4b TYPE7；內部弧光耐受能力(IEC)需依照IEC 61641標準製作，至少需能耐受100KA 0.3sec；主開關及分路開關使用ACB，Mian以及Tie ACB使用額外保護電驛裝置；每個分路開關需作並聯抽出型之開關底座，以作為不斷電之保養維護。
 - 3. 提供特別的工具及附件，以完成安裝及維修工作。
 - 4. 樓板軌道槽鐵及配線所需之BUS RISER。
 - 5. 自接地箱至低壓電盤之接地配線。
 - 6. 自直流電源分電盤至本工程內之低壓電盤所需之直流電源引進點間之配線。
 - 7. 每一變電站提供兩台ACB用維修台車，台車須符合現場環境操作使用。
 - 8. 需負責本包工程範圍內之開孔及防火填塞。
 - 9. 多功能電錶需有RJ45通信與DI至少一點以上。
 - 10. 變電站內低壓分路引出之初期未使用地方需施作封蓋。
 - 11. 預留接線端子供將來安裝自動功率因數調整器(APFR)用。(Facility Substation)

12. MCB狀態點需配線至METER並回傳SCADA作為告警用。
13. 本工程範圍內之設備器材必須為出廠一年內新品。
14. 每一變電站應提供個別容量之Spare ACB一組供保養切換用。
15. 模鑄樹脂密封乾式變壓器，如圖所示。
16. 運送及安裝變壓器於指定之箱體內。
17. 配合送電前所需之機電相關測試、測試報告等，並提供給負責台電報竣之相關人員，送電後需進行變壓器PD檢測。
18. 承商需協助整合竣工圖、控制等相關竣工圖面。
19. 承商需提供銅鋁合片與避震墊。
20. 提供REVIT 3D圖檔。
21. 變壓器箱箱內之月BUS，軟銅帶及固定另件。
22. 備援變壓器至單元變電站變壓器間的6000A Busway銜接。
23. 單元變電站的長X寬X高限制如下：
CUB3 480V Facility Substation 17m(L)x2m(W)x3m(H)
FAB 480V Facility Substation 9.4m(L)x2m(W)x3m(H)
FAB 480V Tool Substation 10m(L)x2m(W)x3m(H)

1.2 適用的規範

- A. SECTION 16260-A - LV VOLTAGE SWITCHBOARD
- B. SECTION 16270-A - CAST RESIN ENCAPSULATED DRY TRANSFORMERS

1.3 送審

A. 投標者之送審

1. 每一組配電盤之材料、設備、顏色及零件表。
2. 製造廠資料：包含所有組件之規格說明及主要設備如低壓電力空氣斷路器、比流器、比壓器等之完整的型錄及記錄資料。

3. 繪製配電盤之組立圖、基座圖、平面圖、前視圖、後視圖及側視圖，但只需繪製一組典型之圖面為代表即可。

B. 承包商之送審

對所有設備組件與零配件，承包商將提供以下之資料：

1. 三線圖與結線圖：三線圖顯示出所有儀表用變壓器、電驛、儀表、儀表開關，以及其他相關的裝置。結線圖則顯示包括控制、保護和計量的情形。
2. 配線圖應顯出端子板位置，以及每個單元內之配線和各單元間彼此交連配線情形。
3. 一般配置圖（general arrangement drawings）包括下列各項：
 - a. 配置圖〔包括前視圖（FRONT VIEW），平面圖（頂視圖）、側視圖（elevation drawings）〕以及底視圖（BOTTOM VIEW）。
 - b. 導管和電纜進線口位置。
 - c. 匯流排條位置和組態（configuration）。
 - d. 電纜進線及出線端接之位置。
 - e. 基礎螺栓（anchor bolt）之位置。

1.4 廠商製造原則：

製造過程限於廠內製造，不得轉包。

- END -

SECTION 16260-A LV VOLTAGE SWITCHBOARD

PART 1 - GENERAL REQUIERMENTS

1.1 DESCRIPTION

This section describes the general rules to guarantee maximum level of quality and performance for a low voltage switchboard including all stages of manufacture, furnish, install and test all equipment, accessories and materials required for the installation low voltage Switchboards in accordance with specifications and drawings. This technical specification covers the low voltage switchboards, which is a type tested switchgear assembly (TTA) .Type tests serve as a proof of the characteristic properties of a particular switchgear design. Their purpose is to prove that the stipulated constructional rules have been maintained. Type tests have to be carried out on a sample of a particular TTA or on part of the TTAs that are manufactured in series in the same or similar form and equipped in the same or a similar way.

1.2 APPLICABLE CODES AND STANDARDS

- A. The work shall be carried out in accordance with this specification, the contract drawings and the standards listed hereunder.
The following codes and standards provide an acceptable level of quality for materials and products:
 - 1. IEC 61439-1&2 : Low voltage switchgear and controlgear assemblies
 - 2. IEC 60529: Degrees of protection provided by enclosure (IP Code)
 - 3. IEC 60947-3: Low-voltage switchgear and controlgear - Part 3 :
Switches, disconnectors, switch-disconnectors and fuse-combination units
 - 4. IEC 61641: Enclosed low-voltage switchgear and controlgear assemblies
- Guide for testing under conditions of arcing due to internal fault.

1.3 QUALITY ASSURANCE

- A. All materials and products shall be new, sound and uniform in quality, size, shape, color and texture and shall be free from defects.
- B. The contractor shall be responsible for ensuring that the required standards of quality control as mentioned in relative sections are maintained for the proposed switchboards.
- C. If requested by the client, the supplier shall provide proof of application of a quality procedure complying with standards. This means:

1. use of a quality manual approved and signed by a management representative,
 2. regular updating of this manual so that it reflects the most recent applicable quality control procedures,
- D. The manufacturer must have a sound experience in the field of LV switchboards for not less than ten years, and should have a presentable reference list of the already supplied equipment of the same type and same make and that this equipment has been in satisfactory operation.

1.4 SUBMITTAL

- A. Shop Drawings and Manufacturer's Data:
The contractor shall submit to the client/client representative for review, detailed dimensioned shop drawings and manufacturer's literature where deviations from the contract drawings or specification exist. Shop drawings and/or data sheets shall be based on information stated in the specifications and as shown on the contract drawings and shall show all pertinent deviations and data for the fabrication and complete installation.
- Manufacturer's data sheets shall be submitted indicating the necessary installation dimensions, weights, materials and performance information. The above information may be provided by standard catalogue sheets marked to indicate the specific items provided.
- B. Operation and Maintenance Instructions:
The contractor shall furnish data covering model, type and serial numbers, capacities, maintenance and operation of each major item of equipment or apparatus in accordance with the requirements of the contract documents. Operating instructions shall cover all phases of control.
- C. Test Reports:
The contractor shall submit test reports to the client/client representative for review.
- D. Recommended Spare Parts List:
The contractor shall submit recommended spare parts list to the client/client representative for review.
- E. As Built Drawings:
The contractor shall submit the as built drawings to the client/client representative for review and approval.
- F. Short circuit calculations proofing the right selection of all components as per site conditions along with coordination study results to be used as protection setting for all protection devices to achieve the discriminations concept

1.5 COORDINATION

- A. The contractor shall be held responsible for the proper coordination of all phases of the work under this contract.

- B. It shall be the responsibility of the contractor to coordinate the work and equipment as specified herein with work to be performed and equipment to be furnished under other sections of the specifications in order to assure a complete and satisfactory installation.

1.6 PRODUCT DELIVERY, STORAGE AND HANDLING

- A. Deliver, store, protect and handle products to site as per manufacturer's instructions.
- B. Store switchboards in clean and dry space. Inspect for damage. Maintain factory wrapping or provide an additional heavy canvas or heavy plastic cover to protect units from dirt, water, construction debris and traffic.
- C. Lift only with lugs provided for the purpose. Handle carefully to avoid damage to switchboard internal components, enclosure and finish.

PART 2 - PRODUCTION

2.1 TECHNICAL CHARACTERISTIC

- A. The switchboards shall be designed and tested for the following electrical values:
 - 1. Rated insulation voltage: 1000 V AC
 - 2. Rated max. operating voltage: 690V AC
 - 3. Rated operating voltage: 480V
 - 4. Frequency: 60 Hz.
 - 5. Insulation level: 1 KV- 60 Hz
 - 6. Impulse withstand voltage: 8 KV peak
 - 7. Busbar rated current As shown on the drawings.
 - 8. Short-circuit current As shown on the drawings
 - 9. Internal arc withstand current : 100KA, 0.3 sec
- B. Switchboard Mechanical Characteristics
 - 1. Forms of separation 4b
 - 2. Protection degree IP 22 for indoor application
 - 3. Bus bar material Cu. , Bare
 - 4. Impact strength (IK) $\geq$ IK8

2.2 SERVICE CONDITIONS

- A. The switchboard shall be suitable for operations at a height of less than 2000 meters above sea level.
- B. The switchboards shall be capable of operating normally within the following temperature range
 - 1. Maximum air temperature: + 40 ° C
 - 2. Minimum air temperature: - 5 ° C

- C. Manufacturer shall declare whether switchboard is able to operate in air temperature higher than + 40 °C and if current derating is necessary.

2.3 CONSTRUCTION AND OPERATION

- A. The low voltage switchboard shall have a permissible asymmetrical short circuit current up to short circuit current shown on the drawings for 1 second. The busbars shall be designed for mounting on insulated supports that are sufficient in number to accept the electro-dynamic forces resulting from the flow of the peak asymmetrical short circuit current.
- B. The low voltage switchboard shall have an earthing circuit including a bar that can be removed for isolation purposes during the necessary insulation measurements (removal of the bar shall require a tool)
- C. Cable entry shall be via the bottom or the top.
- D. Natural ventilation shall make it possible for the Switchboard components to operate within the recommended temperature ranges.
- E. The system shall make it possible to implement fixed or withdrawable distribution sections, positioned side by side, which together form an assembly referred to as an electrical switchboard.
- F. Selection of switchboard components shall be made in compliance with standard IEC 60947 and related specification in the contract documents.
- G. The low voltage electrical switchboard shall be made up of identified functional volumes including the busbar compartment, switchgear and controlgear component compartment, connection compartment and auxiliaries compartment.
 - 1. The different busbar shall be the main busbars, distribution busbars and auxiliary busbars.
 - 2. The main busbars shall be made of electrolytic copper.
 - 3. The compartment shall be located inside a metal enclosure with walls providing protection against direct contact with live parts and guaranteeing a degree of protection (IP) requested. The frame, the external panels (doors, side and rear panels, tops) and internal elements (ducts) shall be made of folded metal steel sheet and protected by epoxy polyester powder paint coating.
- H. The short circuit current rating of the system shall be determined by the available fault current at the Low Voltage Switchgear. All interrupting ratings shall be accomplished without the use of limiter fuses.
 - 1. Withdrawable circuit breakers shall have four different positions:
 - 2. Connected
 - 3. Test
 - 4. Disconnected
 - 5. Removed
- I. The switchboard shall be suitable for installation side by side and capable of receiving lateral ducts for busbar, cables and terminals.

- J. The switchboard cover panel shall be removable.
- K. The construction system shall provide a complete set of elements for installing fixed or Withdrawable switching and protective devices, measurement devices and control/ monitoring devices in the switchboard.
- L. The switchboard shall be certified to guarantee protection against internal arcing by reaching arc withstand values up to 100kA for 0.3s.

2.4 CONNECTIONS

- A. The design of the switchboard shall be such that it will be possible to enter the power cables or busbars trunkings with standard tested accessories by the top, bottom (cables) without modifying busbar location or switchboard height.
- B. All connection points shall be totally accessible. Upper and lower (when cubicles are equipped with plinths) cross-members shall be removable to make easier the entry of large size cables.
- C. The main supply shall be performed by cables, in an area dimensioned according to the number and cross-section of the cables.
- D. The top plates shall be equipped with suitable devices in order to maintain the degree of protection. The useful surface shall be largely dimensioned. The cables shall be connected to specific bar tails or terminals. A flanging device shall be provided to prevent stress on the connections and support the electrodynamic strengths related to the prospective short-circuit current of the switchboard. The switchgear mounting plates shall be equipped with adjustable cable fastening tabs.
- E. All LV signal/control wiring shall be carried out in 600/1000 V grade insulated multi stranded copper conductor and shall not be smaller than 1,5 sq. mm. Connection shall be done in the lateral connection compartment or at the top of the column on terminal blocks with spring

2.5 CURRENT DISTRIBUTION

- A. The rating of the main busbar shall be 6500 A maximum.
The main busbars shall be placed at the top or at the bottom of the columns. It shall be composed of rectangular copper bars, The busbars shall be of constant section according to the nominal current.
The bars shall be braced by means of reversible insulating supports fixed to the frame, designed for both 5 or 10 mm thick bars, in number according to the short time current I_{cw} and the column width. The configuration shall have been designed and tested according to IEC 61439-1&2.
The thermal dissipation shall be carefully studied; large busways without influence of one phase to each other shall be preferred. For electromagnetic radiation limitation, neutral bar shall be located together with phase bars, placed in the front to securitize maintenance interventions.

- B. Connection of the main busbars shall use joints arranged in order to make all necessary operations without being obstructed by the power cables. They shall be sliding to avoid any bar drilling and equipped with torque nuts. Switchboard extension shall be possible at each end by adding columns with voltage off. The main busbars shall have a standardized position and phase order to simplify the design and increase the safety.
- C. The distribution busbars shall be installed in a compartment placed right or left side of the switchgear one. It shall allow the connection at any height of feeders installed here and there without any bolts or drillings. Its design shall provide compactness, lightness, improved natural convection, by using channeled aluminum bars. To insure a good electrical contact, aluminum shall be covered with a deposit of high velocity projected copper, on all the height of the contact side of the bar. The other sides of the bar shall be anodized.
The bars shall be braced by means of insulating supports in number and distance between axis according to the short time current I_{cw} . The bars layout shall authorize the front access of any connection point. The configuration shall have been tested according to IEC 61439-1&2.
Links without drilling between main and distribution busbars shall be preferred. Their tightening shall be ensured by torque nuts.
- D. The installation system shall allow functionalized sub-distribution groups consisting of modular switchgear rows. The busbar shall be fully insulated with clipped on covers to ensure a IPxxB protection against direct contacts and for easy maintenance.
Prefabricated connections are recommended with torque nuts or spring terminal to ensure the correct tightness and sufficient distribution capacity in optimized space. Prefabricated distribution components should have well distributed vents to ensure natural convection and optimum cooling of current conductors.

2.6 FUNCTIONAL UNIT DATA

- A. All functional units of the same modularity shall be interchangeable. All accesses to the functional units shall be possible from the front.
- B. The mounting devices shall be equipped with guides and marks for easy positioning of the switchgear.
- C. The devices shall be fitted to the mounting plate by screws, without nuts to avoid any accidental fall of nuts in the equipment.
- D. The front plates shall be fitted to a reversible swiveling frame giving easy access to the switchgear during interventions.

2.7 PROTECTION AND SAFETY

- A. The low voltage electrical switchboards shall ensure the safety of life and property as well as provide a high level of continuity to service
 - 1. Switching safety shall be ensured by mechanical device preventing on load withdrawal

2. Operating safety shall be ensured by compartmenting in compliance with standard IEC 61439-1&2 and according to form types 4b.
 3. Current interruption shall be of the “visible break isolation” or “positive contact indication” type as defined by standard IEC 60947-3.
- B. In view of reducing the risk of electrical shock
1. Power and control circuit shall be separate and completely isolated.
 2. Auxiliary circuits shall be of the extra low voltage type.

2.8 DESIGN & ROUTINE VERIFICATIONS:

Standard IEC 61439- 1&2 clearly defines the type of verifications (design verification and routine verification) that must be conducted by both organizations involved in final conformity of the solution: the Original Manufacturer guaranteeing “assembly system” design and the Assembly Manufacturer who is responsible for the final conformity of the LV switchboard.

All the verifications proposed by standard IEC 61439- 1&2 contribute to the achievement of 3 basic goals: safety, continuity of service and compliance with end user requirements.

To guarantee consistency of equipment installed, the “Assembly system” and the devices will be supplied by the same manufacturer

2.8.1 Requirements applying to the “original manufacturer”:

To be compliant with standard IEC 61439 -1&2 the original manufacturer carries out the original design and the design verifications for the 10 following main functions that have to be certified through an independent certification body (ASEFA, ASTA, KEMA,...)

Design verifications for 10 following main functions:

No. 1 – Voltage stress withstand capability

Measurement of clearances and creepage distances,

Power frequency dielectric test,

Impulse withstand voltage test when clearances are greater than specified values

No. 2 – Current-carrying capability

Temperature rise tests (The tests have to be made for each circuit alone at 100% and for the complete assembly taking into account the diversity factor)

No. 3 – Short-circuit withstand capability

Short-circuit tests (I_{cc} and I_{cw}) of the main circuit, including the neutral conductor, and of the protection circuit .

No. 4 – Protection against electric shock

IP XXB test and verification of insulating materials

Mechanical operation tests

Verification of dielectric properties

Measurement of the resistance between each exposed conductive part and the PE terminal

Short-circuit strength of the protection circuit

No. 5 – Maintenance and modification capability

IP tests

Mechanical operation tests (especially for removable parts)

No. 6 – Capability to operate the electrical installation

By inspection

Impulse withstand voltage test of isolating distance for optional withdrawable units

No. 7 – Protection of the Assembly against environmental conditions

IP test

IK test

2.8.2 Requirements applying to the “assembly manufacturer”:

To be compliant with standard IEC 61439 -1&2 the assembly manufacturer carries out routine verifications for the 10 following main functions. These routine verifications must be realized for each manufactured assembly.

No. 1 – Voltage stress withstand capability

Visual inspection of clearances (subject to design conditions and creepage distances)

Power frequency dielectric test

No. 2 – Current-carrying capability

Visual inspection

Random verification of tightness

No. 3 – Short-circuit withstand capability

Visual inspection

No. 4 – Protection against electric shock

Visual inspection of basic and fault protection

Random verification of tightness of the protective circuit connections

No. 5 – Maintenance and modification capability

Effectiveness of mechanical actuating elements

Check protection of persons against electric shocks

No. 6 – Capability to operate the electrical installation

Visual inspection

Effectiveness of mechanical actuating elements and function test (where relevant)

No. 10 – Protection of the Assembly against environmental conditions

2.8.3 Requirements concerning main & sub-main LV switchboard design:

The following design rules must be implemented in order to facilitate construction and in particular maintenance of the installation.

The main & sub-main LV switchboard must be designed to ensure a clear vision of the separation into the 3 volumes below:

- The volume dedicated to switchgear installation,
- The volume dedicated to busbar mounting,
- The volume dedicated to connections and to insertion of feeder cables.

The aim of this LV switchboard architecture is to partition the switchboard into different zones according to the different users:

- volume dedicated to the switchgear => panel-builder and operator,
- volume dedicated to busbars => panel-builder,
- volume dedicated to connections and to insertion of feeder cables => contractor and maintenance operator.

PART 3 - EXECUTION

3.1 WORKMANSHIP

- A. Materials, products and equipment furnished by the contractor, shall be installed and all work shall be performed in a first-class workmanship manner, in conformity with the best trade practices and the printed directions of the applicable manufacturers; by skilled workers equipped to produce satisfactory results; in a safe, substantial manner so as to avoid undue stresses, rigid enough to prevent undue movement, so as to present a neat, orderly appearance and to facilitate operating, servicing, maintenance and repairing.

3.2 FOUNDATIONS AND SUPPORTS

- A. The contractor shall provide concrete pedestals(The construction site has been implemented, and the project must be modified to meet the requirements.), anchor bolts, hangers, channels, saddles, etc., for installation of equipment and apparatus shown on the drawings and specified in the various sections.

3.3 EQUIPMENT ERECTION

- A. All electrical equipment shall be installed in accordance with the manufacturer's recommendations, good electrical engineering practice, and the relevant drawings and specifications.
- B. All metal surfaces to be bolted shall be thoroughly cleaned before assembly. The connections shall be tightened with manual torque wrenches to the manufacturer's erection instructions.
- C. Splices shall be implemented to ensure the electrical continuity of the horizontal busbars, auxiliary buses and the proactive conductor between adjacent sections.

- D. It shall be possible to secure the sections to a floor that is flat by anchoring directly to a concrete floor using anchor bolts or by securing to ordinary metal profiles.

3.4 EQUIPMENT TESTING AND COMMISSIONING

- A. After the installation is complete and properly adjusted, the contractor shall conduct operating tests. The various equipment and systems shall be demonstrated to operate in accordance with the requirements of the contract documents. Tests shall be performed in the presence of the client/client representative. The contractor shall provide electric power, instruments and personnel necessary for performing the various tests.
- B. The testing of all electrical equipment shall include, but not be limited to, the items below. This shall be in addition to testing specified elsewhere in this specification.
 - 1. General equipment check.
 - 2. Field wiring and ground system verification.
 - 3. Equipment adjustment.

- END -

SECTION 16270-A CAST RESIN ENCAPSULATED DRY TRANSFORMERS

PART 1 – GENERAL

1.1 DESCRIPTION

This specification covers designing, furnishing, and installing and testing of:

- A. 4000/5000 KVA, AN/AF, 22.8KV – 492/284V, 60HZ, three-phase two winding cast resin encapsulated dry transformers.

1.2 APPLICABLE CODES AND STANDARDS

The standards and codes applicable to only a portion of the work specified in this specification are referenced in the relevant parts or clauses. Standards and codes which are generally applicable to the work of this specification are listed below:

- A. IEC 60076-11 Power transformers dry-type transformers
- B. CNS – Chinese national Standard 13390 C4468 Dry Type Transformers

Where references are made to standards, the Contractor may submit equipment built and tested according to other standards providing that the requirements of the referenced standards are complied with as a minimum.

The year of revision of each standard and code is given to facilitate references. Nevertheless, the revision current at the date of tender shall be the basis of the contract.

If there is any conflict between this specification and any reference standard or code, the Contractor shall inform the Owner and ask for clarification before manufacture.

1.3 SUBMITTALS

The Contractor shall submit the following for all equipment components and accessories:

- A. Data: Transformer specification sheet as similar to the appendix B of ANSI C57.12.00 - 1980.
- B. Certificates and Reports: Certification that the transformers will meet the test of IEC 60076-11.
- C. If aluminum windings are used for transformers, the Contractor shall submit the formula to calculate the temperature rise and short circuit current density in order that the short circuit temperature rise of aluminum winding will be limited to 200°C.

1.4 WORK INCLUDED

- A. Furnish the cast resin encapsulated dry transformers.
- B. Follow local government regulations and get approval of energizing all transformers.

PART 2 – PRODUCTS

2.1 SERVICE CONDITIONA

- A. Altitude
A height above sea-level not exceeding 1000m.
- B. Temperature of cooling air
Never exceeding 40°C and never below 0°C. In addition, an ambient air temperature never exceeding the following values:
 - 1. 33°C average in any one day.
 - 2. 20°C average in any one year.
- C. Wave-shape of supply voltage
A supply voltage of which the wave-shape is approximately sinusoidal.
- D. Symmetry of polyphase supply voltage.
For polyphase transformers, supply voltages which are approximately symmetrical.
- E. The transformers must be provided repairs serving and protecting firmly in 5 years.

2.2 DESIGN REQUIREMENTS

- A. The transformers shall be vacuum impregnated or machine roving cast-resin encapsulated, dry-type, three-phase, 60 HZ, with at least class F, 155°C insulation for high-tension and low-tension side. Transformer shall be rated for continuous operation at rated KVA with a maximum winding temperature rise of 95°C and shall be enclosed in ventilation steel enclosures. Forced air cooling shall be provided, such that the transformer rating can be increased by at least one third without exceeding the specified temperature rise above the specified maximum ambient temperature.
- B. The transformers primary tapping is 22.8KV, secondary voltage is 492V or 213V at no load and shall be provided with two 2-1/2 percent full capacity taps above rated primary voltage. Taps shall be accessible from front or rear of unit.

- C. Transformer impedance shall be as shown with NEMA Standard Tolerance.
- D. Transformer shall be mounted on approved vibration and noise isolating and damping supports. Core and coil assemblies shall be rigidly braced to withstand mechanical forces under line fault conditions and to resist vibration and shock forces during shipment.
- E. Each high voltage winding and each low voltage winding of each phase shall be separately cast as a rigid tubular coil of copper or aluminum conductors. High voltage and low voltage windings shall be coaxial with no mechanical connection. The windings shall not absorb moisture and shall be suitable for both storage and operation in adverse environments, including prolonged storage in 100% humidity at temperatures ranging from 0°C to 40°C and shall be capable of being energized immediately after such storage without predrying.
- F. The transformers shall have a delta primary – grounded wye secondary phase configuration unless noted otherwise. The phase relationship shall be standard: HV to lead LV by 30°.
- G. All transformer winding terminations shall be provided with NEMA CC 1 Standard copper connectors. Bus bar terminations shall be in accordance with NEMA standards and supported with insulators not more than 400 mm from the connection holes.
- H. Temperature sensing elements shall be embedded in windings.
- I. The core shall be visibly grounded to the base and the base provided with a grounding connection pad.
- J. The outside surfaces of the core shall be protected against corrosion.
- K. Coils shall be vacuum impregnated or machine roving, and cast-resin encapsulated to give it a tough moisture and dirt resistant overall finish. The coils shall be non- flammable and self-extinguishing when exposed to electric arc during fault conditions as specified by IEC Standards.
- L. The maximum sound pressure level will be 69 dbA, except, measured at 30cm from external surface at AN rating.
- M. Unless noted otherwise, the transformers shall be designed for non-linear loads with K-factor rating of K1.
- N. Transformers will be of environmental class E2 and of climatic class C2 as defined in IEC 60076.
- O. These transformers will be of class F1 as defined in IEC 60076-11. F1 class will be indicated on the rating plate.
This test must have been performed in accordance with IEC 60076-11.

- P. Measurement of partial discharges.
For this measurement, the acceptance criteria will be less than or equal to 5 PC at 1.30 Un.
- Q. The contractor shall provide the material list for transformer manufacturing.
- R. Random wound coils are not acceptable for use in medium voltage windings.
- S. Cooling fans shall be high reliability, capable of operating 5 years continuously without failure.
- T. Transformers shall be suitable for switching by all types of switching devices, i.e. vacuum or SF6 breakers without detrimental effect on the windings.

2.3 RATING

- A. Rated capacity: 4000/5000 KVA, AN/AF.
 - 1. Rated voltage / connection :
 - a. HV 22.8 KV, delta.
 - b. LV 480/277 V, Wye, neutral effective grounded
 - c. Connection Dyn1
 - 2. System insulation classes :
 - a. HV, 22.8 KV 25 KV (ANSI)
 - b. LV 1.1 KV
 - 3. Basic impulse insulation levels :
 - a. HV, 22.8 KV 150 KV
 - 4. Transformers shall be suitable for impulse testing.
 - a. (BIL x 100%) during FAT
 - 5. Power frequency test voltage :
 - a. HV, 22.8 KV 50 KV
 - b. LV 3 KV
 - 6. Impedance 8 %

2.4 SHOP TESTING

- A. General requirements
 - 1. Tests shall be made at the manufacturer's works, unless otherwise agreed between the Owner and the Contractor.
 - 2. Dielectric tests in accordance with square-source voltage withstand test, induced overvoltage withstand test and lightning impulse test shall be made with the transformer at approximately ambient temperature.
 - 3. All external components and fittings that are likely to affect the performance of the transformer during each test shall be in place.

4. Tapped windings shall be connected on their principal tapping unless agreed by the Owner otherwise.
 5. The test basis for all characteristics other than insulation is the rated condition.
 6. The reference temperature to which the load loss, impedance voltage and short-circuit impedance are to be corrected shall be the maximum winding temperature rise plus 20 K. If transformers have windings of different insulation system temperatures only one reference temperature is to be used, that is, the one relating to the winding having the higher insulation system temperature.
- B. Routine tests: Tests shall be performed in accordance with IEC 60076-11, and shall include as next:
1. Visual & dimensions.
 2. Measurement of winding resistance.
 3. Measurement of voltage ratio and check of vector relation.
 4. Measurement of insulation resistance.
 5. Measurement of impedance voltage and load loss.
 6. Measurement of no-load loss and current.
 7. Separate-source voltage withstand.
 8. Induced over voltage withstand test.
 9. Partial discharge measurement, after install and energize must be re-measurement.
- C. Type tests: Tests shall be performed in accordance with IEC 60076-11 and shall include the following type tests:
1. Measurement of insulation Power factor
 2. Lightning impulse test.
 3. Load temperature rise test.
 4. Sound level measurement.
- D. Sound level: The measurement of sound level shall be performed in accordance with IEC 60076-10.

2.5 SPECIAL REQUIREMENT

- A. The transformers will be bolted to the floor. They shall withstand seismic force in accordance with Taiwan Building code 2011 for continued operation with the following factors:
Occupancy factor: $I_p = 1.5$

PART 3 –EXECUTION

3.1 FIELD TEST

After the equipment has been installed, inspected, and placed in operating condition, it shall be field-tested. The field test shall demonstrate that the

equipment and components

Function in compliance with the specification over the entire range of operation.
Required field test includes,

All standard tests recommended by factory as part of commissioning plus following tests.

A. Visual and Mechanical Inspection:

1. Check primary, secondary, and ground connections.
2. Clean and inspect bushings.
3. Inspect bushing clamps and gaskets.
4. Check accessory devices for condition and proper operation.

B. Electrical Tests:

1. Insulation Resistance Test & Polarization index test: Megger transformer windings high to low and ground, low to high and ground, and high and low to ground.
2. Perform High potential test
3. Perform turns ratio test.
4. Where auxiliary cooling has been provided, verify proper operation of such equipment.

C. Test transformer according to IEC standards.

- END -